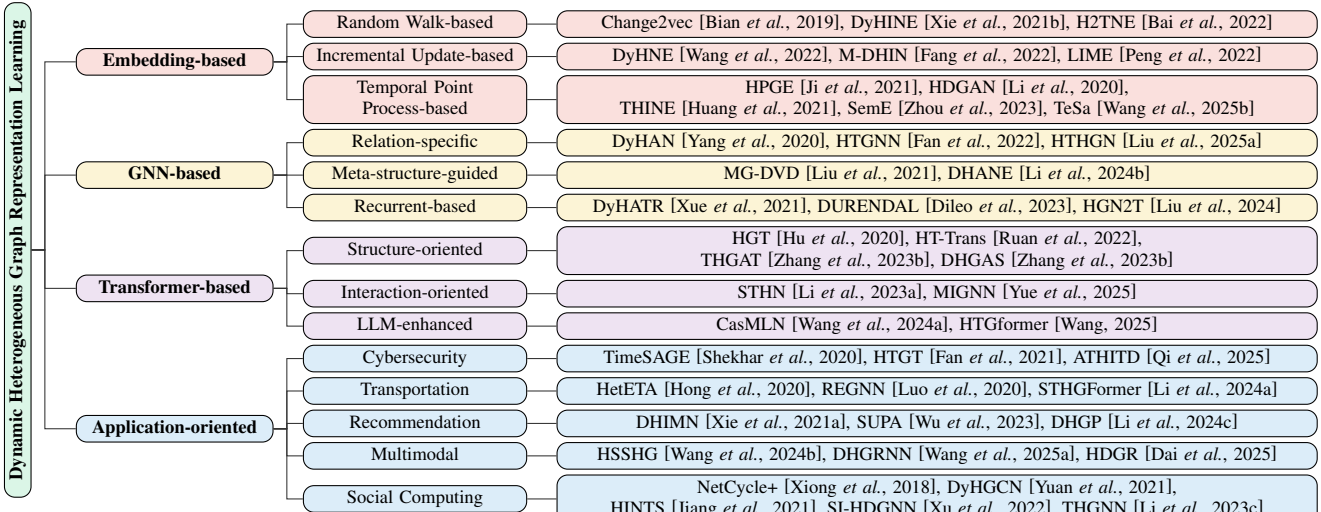

Dynamic Heterogeneous Graph Representation Learning
Embedding-based
Random Walk-based
Change2vec [Bian et al., 2019], DyHINE [Xie et al., 2021b], H2TNE [Bai et al., 2022]
Incremental Update-based
DyHNE [Wang et al., 2022], M-DHIN [Fang et al., 2022], LIME [Peng et al., 2022]
Temporal Point Process-based
HPGE [Ji et al., 2021], HDGAN [Li et al., 2020], THINE [Huang et al., 2021], SemE [Zhou et al., 2023], TeSa [Wang et al., 2025b]
GNN-based
Relation-specific
DyHAN [Yang et al., 2020], HTGNN [Fan et al., 2022], HTHGN [Liu et al., 2025a]
Meta-structure-guided
MG-DVD [Liu et al., 2021], DHANE [Li et al., 2024b]
Recurrent-based
DyHATR [Xue et al., 2021], DURENDAL [Dileo et al., 2023], HGN2T [Liu et al., 2024]
Transformer-based
Structure-oriented
HGT [Hu et al., 2020], HT-Trans [Ruan et al., 2022], THGAT [Zhang et al., 2023b], DHGAS [Zhang et al., 2023b]
Interaction-oriented
STHN [Li et al., 2023a], MIGNN [Yue et al., 2025]
LLM-enhanced
CasMLN [Wang et al., 2024a], HTGformer [Wang, 2025]
Application-oriented
Cybersecurity
TimeSAGE [Shekhar et al., 2020], HTGT [Fan et al., 2021], ATHITD [Qi et al., 2025]
Transportation
HetETA [Hong et al., 2020], REGNN [Luo et al., 2020], STHGFormer [Li et al., 2024a]
Recommendation
DHIMN [Xie et al., 2021a], SUPA [Wu et al., 2023], DHGP [Li et al., 2024c]
Multimodal
HSSHG [Wang et al., 2024b], DHGRNN [Wang et al., 2025a], HDGR [Dai et al., 2025]
Social Computing
NetCycle+ [Xiong et al., 2018], DyHGCN [Yuan et al., 2021], HINTS [Jiang et al., 2021], SI-HDGNN [Xu et al., 2022], THGNN [Li et al., 2023c]